```
function liveness (marking)
  local markings   = reachable (from)
  local components = tarjan (markings)
  for _, transition in ipairs (transitions) do
    transition.liveness = {
      l0 = true,
      l1 = false,
      l2 = false,
      l3 = false,
      l4 = true,
    }
    for marking in pairs (markings) do
      if marking [transition] then
        transition.liveness.l0 = false
        transition.liveness.l1 = true
      else
        transition.liveness.l4 = false
      end
    end
    for _, component in ipairs (components) do
      for marking in pairs (component) do
        if  marking [transition]
        and marking [transition].tarjan.component == component then
          transition.liveness.l2 = true
        end
      end
      transition.liveness.l3 = component.is_final and transition.liveness.l2
    end
  end
  return transitions
end
```